Escola d'Enginyeria de Telecomunicació i Aeroespacial de Castelldefels

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

Libraries Documentation

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1. Introduction

In this document, I will compare and analyze which LoRa library is the best for this project. As I want to determine all the possible parameters, I will choose the library that allows the user to select and change more parameters.

All this analysis is considering the use of ESP32-S3 microcontrollers, utilizing the Arduino language and its related libraries.

2. Comparison Between Libraries

In this document, I will compare and analyze which LoRa library is the best for this project. As I want to determine all the possible parameters, I will choose the library that allows the user to select and change more parameters.

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Library	Freqiency (Hz)	Spreading Factor	P_{tx} (W)
Unnoficial Heltec [1]	☒	☐	
Arduino LoRa [2]	16.6	104	
RadioLib [3]		24.7	

Table 1: Library Table Comparison

Bibliography

[1] R. Pihlakas, “Heltec ESP32 LoRa V3.” [Online]. Available: https://github.com/ropg/heltec_esp32_lora_v3

[2] S. Mistry, “Arduino-LoRa.” [Online]. Available: https://github.com/sandeepmistry/arduino-LoRa

[3] J. Gromeš, “RadioLib: Universal wireless communication library for Arduino.” [Online]. Available: https://github.com/jgromes/RadioLib